

5G Technology to Benefit Cyber Physical Systems



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According to a press release by Telecom Regulatory Authority of India (TRAI), the number of mobile phone subscribers in the country has crossed 952.34 million, as of January 2015, and is continuously growing. The Ericsson mobility report from November 2014 also predicts that 45 per cent of Indian telecom subscribers will be covered by fourth-generation (4G) long term evolution (LTE) by the year 2020.

With the introduction of affordable and 4G-supported devices like Xiaomi Redmi Note, Elephone P3000, Lenovo A6000 and Gionee P6, 4G technology has already become accessible to the common man. Where will 4G go next, and what exactly is 5G?

Advances in voice over LTE

Modern-day phones perform a lot more than the traditional function of voice communication. These features were enabled in 2G and 3G networks through voice over Internet protocol (VoIP). Though it has a wide user base and is utilised by popular apps like WhatsApp, Skype and Google Hangouts, the quality of service components is not the best. Packets of data are sent over the Inter-

net and a portion of these packets is often dropped under stress. This is where voice over LTE (VoLTE) comes into play.

Being service-driven, VoLTE works with 4G LTE network and uses IP multimedia subsystem (IMS) and radio access network (RAN) features. The IMS layer offers low latency, error correction and several features for better quality of service. VoLTE service currently available in India has to improve considerably, points out Srinivasa Moorthy, director of design engineering, D4X Technologies Pvt Ltd. Packets sent using VoLTE sometimes do not suffer the same delay in a sequence. The network should be capable of minimising this delay and autocorrecting it.

Small cells deployment

Vodafone and advertising firm JCDecaux signed an agreement for a global small cell rollout a few months back. You may ask, what does an advertising firm have to do with mobile network technology? Under the deal, small cells will be placed in street furniture and billboards to improve network performance in congested urban areas. Talk about creative solutions!

Small cells basically take mobile signals transmitted by macrocells and direct these to mobile transceiver units within a range of 10 to several hundred metres, and vice versa. These have gained much importance in LTE era and are usually deployed both outdoors and indoors, where signals are weak or networks are overloaded. These areas include rural regions, basements and parking areas of buildings, home and office spaces.

Unlike a macrocell, a small cell occupies only a small space, emits clean and environment-friendly signals, does not require mobile tower leasing cost and significantly expands the network coverage at any place.

Since these are installed at lower heights, the distance between user terminals

